Zero Birefringence Condition in Lithium Niobate on Insulator Rib Waveguides

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Birefringence, quantified as the maximum effective index difference of two orthogonally polarized modes, is one of the key characteristics of Lithium Niobate on Insulator (LNOI) waveguide devices. As LNOI optical properties strongly depend on the polarization of the propagating light beam, achieving the zero-birefringence condition (ZBC) is highly desired. In this work, a rib LNOI waveguide is designed, and the influence of geometrical dimensions on the birefringence is characterized. From the result, the slab's height is found to be the dominant geometrical factor that significantly contributes to the birefringence, with an optimum slab's height found at 50 nm. For this optimized value, the influence of width and wavelength on birefringence are characterized. The analysis is extended to investigate the properties of group indices that can result in the temporal walk-off phenomenon. This phenomenon can significantly impact the performance of single-photon devices, as it can cause a polarization mismatch between the incident photon and the polarization properties of the medium, resulting in reduced efficiency of devices such as single-photon detectors. In the analysis, the zero birefringence and walk-off are achieved at narrow waveguide width, w of 250 nm and wavelength, $\lambda = 1500$ nm, which is within the telecommunication region band.

1. Introduction

Over the last few decades, photonic integrated circuit (PIC) technology has grown rapidly by fostering both fundamental research of quantum phenomena and a wide range of disruptive quantum technologies. This quantum-based technology offers extensive applications in various fields, including optical communication,[1,2] sensing[3] as well as fast computing[4] by manipulating the properties of light. Silicon on insulator (SOI), Indium Phosphide (InP), Silicon Nitride (SiN), and Lithium Niobate (LN) are well-known materials used in the development of this technology. The latter is highlighted as a prominent material due to its inherent properties, including strong second-order nonlinearity, wide transparency windows, and small index difference,[5,6].These outstanding material properties tremendously impact the development of on-chip integrated photonic technology, making the promise for LN devices to eventually replace the bulky conventional off-chip systems. On top of that, the LN ability to inherently generate entangled photon pairs makes it a prominent material for quantum information applications such as quantum cryptographic systems,[7] teleportation[8] and entanglement swapping.[9]

Despite these unique properties, LN suffers from the hard dry etching process due to chemical instability that leads to difficulties of dimension shrinkage.[10,11] Its non-uniform surface roughness results in an increased scattering loss. The recent emergence of Lithium Niobate on Insulator (LNOI) has recuperated the interest in LN, holding a promise for ultra-compact, and high-efficacy complex integrated circuit technology.[12] In most of the PIC applications, this non-centrosymmetric crystal is exploited to support high-efficiency nonlinear optical processes such as second harmonic generation or parametric processes like difference frequency generation or four-wave mixing. The latter process is widely applied for the generation of photon pairs via the interaction of high energy pump laser with the nonlinear crystal.[13] This spontaneous process results in the creation of two lower-energy photons (signal and idler photons) on the condition of energy and momentum conservation. Eventually, the momentum conservation requirement is satisfied when the wave vectors of the signal and idler photons add up to the wave vector of the original pump photon. However, in most cases, the crystal structure introduces a refractive index difference between the signal and idler photons resulting in a phase mismatch. It is important to eliminate this phase mismatch prior to exploiting the nonlinearity to obtain high spectral purity of photon pairs. Several methods

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were proposed to fulfill the phase matching condition, including modal birefringence control. Small modal birefringence corresponds to better phase matching of two orthogonally polarized modes across the waveguide.

Recently, a number of research works have been reported discussing the zero birefringence condition on SOI waveguides[14] and LNOI waveguides.[15,16] The latter has attracted much attention due to higher intrinsic birefringence as well as stronger nonlinearity. While Schollhammer reported the zero modal birefringence for ridge and buried waveguide structures,[15] we have demonstrated fulfilling the zero birefringence condition (ZBC) on rib structure in this work. To date, a great number of LNOI-based rib photonic structures were fabricated, including ring resonators,[17] modulators[18], and wavelength converters.[19] However, the ZBC for field modes with *x*-propagation direction in *y*-cut LNOI rib structures has been barely discussed previously.[20] The existence of a slab layer underneath the core, which may significantly contribute to stronger mode confinement, was not considered.

In this paper, we perform a mode analysis of rib LNOI waveguide structures using the finite element method in the COMSOL Multiphysics software. From full-vectorial modeling of the waveguide structure, we identify the suitable waveguide geometry, which leads to a modal birefringence-free propagation over a broad spectral range from 600 to 1800 nm. We highlight the geometrical influence of the slab's height as well as the width of the core on the modal birefringence. The analysis is extended to discuss the group indices trends in ultra-compact waveguides, as it may possibly raise the waveguide dispersion. Our outcomes may serve as a fundamental framework for the fabrication process optimization and selection of the suitable wavelength for frequency converters dedicated to, e.g., correlated photon pairs or frequency doubling.

2. Design and Simulation

The mode propagation in anisotropic optical waveguides is determined by the material properties and the geometrical structure. Modal birefringence, which depends on these features, is one of the crucial parameters that determines the functionality of waveguides. In order to fulfill the zero birefringence condition (ZBC) in LNOI waveguides, optimization of their geometry is essential. Here, it is performed for a *y*-cut rib LNOI waveguide whose cross-sectional view is depicted in **Figure 1** The waveguide consists of 300 nm thick LN deposited on top of the 2 μm thick silica (BOX) layer. The existence of this BOX layer minimizes electromagnetic field penetration into the LN substrate. To access negative uniaxial properties, we consider wave propagation along the *x*-axis with TE and TM modes polarized along the *z*-axis (extraordinary) and *y*-axis (ordinary), respectively. Hence, we find slightly lower values of extraordinary refractive indices, n_{eo} compared to the ordinary indices, n_o. The Sellmeier relations are applied to address the dispersive nature of the LN and SiO_2 materials,[21,22] On top of that, the 2 μm conformal coating layer is introduced on top of LN structure, enabling the simultaneous control of diverse optical properties of rib waveguides, hence, allowing stronger modal confinement and great optical guidance. The structure is simulated using the Electromagnetic Waves, Frequency Domain interface in COMSOL Multiphysics, with the maximum meshing size set to 1/5 of the wavelength.

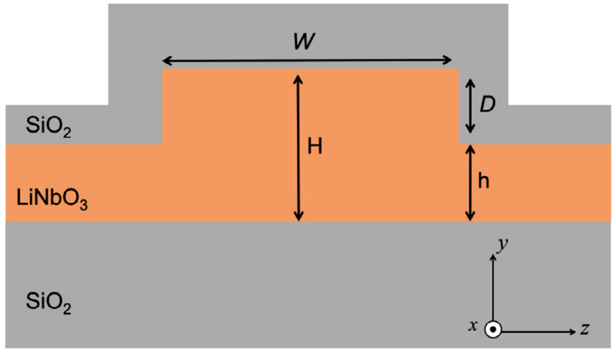

Figure 1. Cross-sectional view of *y*-cut with *x*-propagation LNOI rib waveguide.

We first vary the waveguide width, *w* and perform the simulations for fixed wavelengths of interest to obtain the effective refractive indices, n_{eff} for TE_{00} and TM_{00} modes that correspond to the fundamental eigenmode solution of Maxwell's equations. Unlike the standard refractive indices, n_{eff} provides an average value of the refractive index experienced by a light wave propagating through a medium with non-uniform refractive index distribution. The ZBC is achieved as $n_{eff,TE00} = n_{eff,TM00}$. Continuously, the influence of the slab substrate on the rib structure's birefringence is evaluated by varying the thickness *h* ranging from 50 to 250 nm, which corresponds to a transition of deep to shallow etched waveguide structure.

3. Results and Discussion

3.1. ZBC for a Fixed Wavelength

With the methodology described above, we evaluate the birefringence of the rib waveguide structure. The results are compared with 300 nm thick buried structure in order to characterize the influence of slab thickness on modal confinement and birefringence. **Figure 2** presents the influence of the waveguide's width on n_{eff}. In this case, the height, *H* of the buried waveguide is set at 300 nm with the etched depth, *D* and slab thickness, *h* is maintained at 250 and 50 nm, respectively. At a fixed wavelength, λ of 1000 nm, it is found that the effective indices of the two investigated modes, TE_{00} and TM_{00} are significantly increase as a function of width. Conversely, as the width of the core decreases, the waveguide dispersion intensifies, modifying the effective indices of both TE_{00} and $TM_{00.}$ This ultimately leads to the emergence of crossing points on the effective index curves, as indicated in Figure 2 marking a change in the balance between the mode indices.

Due to the strong optical dispersion inside the waveguide, the effective indices of both orthogonally polarized modes become closer and reach the ZBC point, which is achieved in the buried waveguide at $\{w, n_{eff}\} = \{0.418\ \mu m, 1.700\}$. At this point, both TE_{00} and TM_{00} modes sustain symmetrical mode profiles. For widths of the core beyond this point, the effective refractive index for the TE_{00} mode dominates over that of the mode TM_{00}. Further widening results in stabilized index values, aligning with the

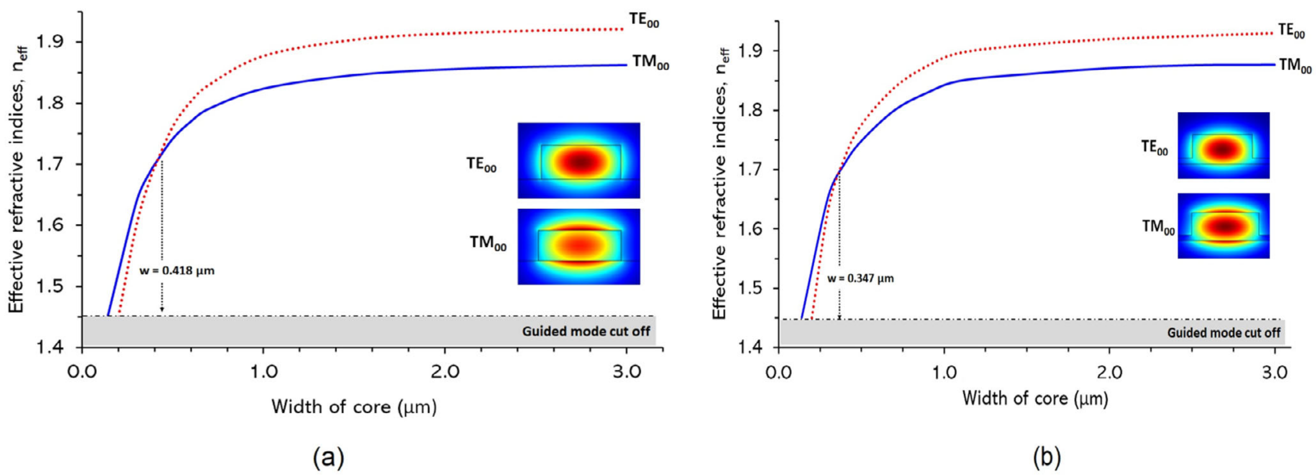

Figure 2. Effective indices of the fundamental mode, TE_{00} and TM_{00} as a function of width at $\lambda = 1000$ nm in a) buried and b) rib waveguide structure with the index overlaps at width, $w = 0.418\ \mu m$ (buried) and 0.347 μm (rib).

conventional trend observed in large modal area waveguides. Interestingly, this ZBC is achieved at $\{w, n_{eff}\} = \{0.347\ \mu m, 1.702\}$ on rib waveguide that is narrower by 0.071 μm compared to buried structure. However, incorporating 50 nm thick slab on rib waveguide marginally increases the n_{eff} by 0.02, contributing to slightly stronger modal confinement. To explore the profound impact of ZBC, we extend the analysis by varying the slab thickness that possibly influences the waveguide dispersion and leakage loss. This finding holds promise for improving the coupling efficiency between the waveguide modes.

Unlike the buried structure, the single mode condition and modal birefringence in the rib structure are strongly dependent on the ratio $r = h/H$ of the slab height, h to overall rib height, H. According to Soref the single mode condition is achievable as the ratio is set in between 0.5 and 1.0, corresponding to shallowly etched waveguides.[23] Under this condition, higher-order modes confined underneath the rib structure are coupled to the outer slab region and yield high propagation loss, leaving only the fundamental mode profile in the rib structure. Numerous research works were reported to highlight the influence of this geometrical relation on the modal properties in SOI,[24–26] as well as LN rib waveguides.[27,28]

The graph in **Figure 3** shows the relation of the zero-birefringence wavelength with the height of the slab, h The graph is separated in the grey region that indicates the large modal area rib waveguide. As we consider a shallow waveguide with $h = 250$ nm, the ZBC is found at a minimum wavelength of 690 nm. The reduction of the slab's height to 150 nm leads to a slight shift of the ZBC wavelength. However, as h approaches 50 nm, the ZBC wavelength is significantly shifted toward longer wavelengths and depends strongly on the height. By reducing the thickness of the slab and increasing the modal area, we thus achieve the single mode condition and stronger mode confinement at a longer wavelength.

Interestingly, at $h = 50$ nm, the second crossing point for the effective indices is found at 1830 nm. As shown in the inset of Figure 3, the spectral region difference that is given by $\Delta\lambda = 830$ nm is found slightly broader compared to the buried structure ($\Delta\lambda = 749$ nm).[15] As expected, we find that the operating wavelength of the LNOI rib waveguide is sensitive to the geometry parameters, and a thinner slab leads to a longer operating wavelength.

Consequently, we present an extended analysis on the commercially available structure of 500 nm thick LNOI waveguide in order to examine its zero birefringence wavelength dependence toward the variation of the slab's thickness. As the width of core is fixed at 400 nm while a minimum slab thickness, h is set at 50 nm, the rib area is expected to double to 0.18 μm^2 compared to a 300 nm thick LNOI waveguide of equivalent slab thickness. This configuration enables enhanced modal confinement within the core structure. As illustrated in **Figure 4**, the computational results reveal that the zero birefringence condition is only achieved as a minimum slab thickness of 100 nm is applied, demonstrating a strong inverse correlation between wavelength and its slab thickness. In this case, the thinner slab enhances the rib waveguide area, leading to a broader effective area for the modal cross-section. The correlation of operating wavelength, mode field diameter (MFD), and effective area is expressed by $A_{eff} = k(\lambda)(\frac{\pi}{4})\ (MFD)^2$. Hence, justifying the need for

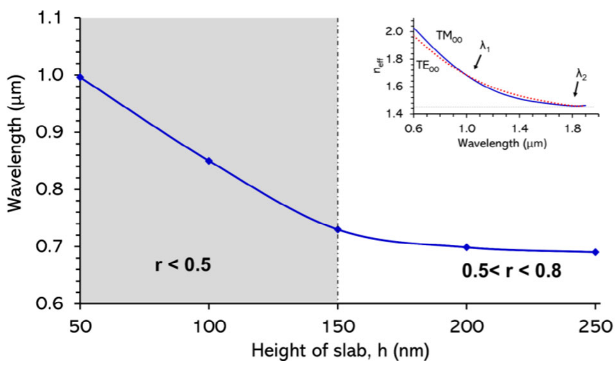

Figure 3. Influence of slab's height, h on zero-birefringence wavelength in the rib waveguide at the fixed width of 347 nm. Inset: Effective indices dispersion for the two investigated modes shows overlapping values at $\lambda_1 = 1000$ nm and $\lambda_2 = 1830$ nm for fixed values of $\{w,h\} = \{347,50\}$ nm.

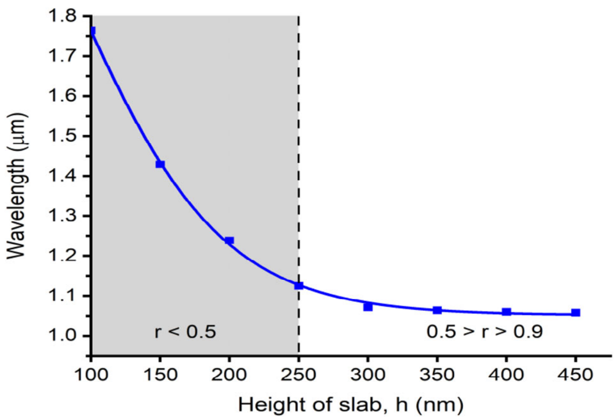

Figure 4. Influence of slab's height, h on zero-birefringence wavelength in 500 nm thick LNOI rib waveguide at the fixed width of 400 nm.

a longer operating wavelength to achieve robust modal confinement in a broader cross-sectional waveguide.

Further increasing the thickness from 100 to 250 nm leads to a significant wavelength down-shifting from 1763 to 1125 nm. However, approaching slab thicknesses greater than 250 nm yields a minimal wavelength down-shifting. In essence, the ZBC in the 500 nm thick LNOI waveguide is notably more sensitive to variations in slab thickness, resulting in a broader wavelength downshifting of 705 nm compared to the previous structure, which achieved a downshifting of 300 nm over a thicker slab. Therefore, precise selection of waveguide dimensions is essential for an efficient operation within the desired wavelength range. Based on this analysis, we anticipate promising wavelength tunability features at $\{H,h\} = \{300,50\}$ nm, which hold significant potential for a wide range of nonlinear applications. Hence, we further investigate these unique features.

3.2. Tuning ZBC in a Range of Wavelengths

While achieving smaller widths is challenging with state-of-the-art techniques, we perform the analysis for widths ranging from 200 to 700 nm at $h = 50$ nm. **Figure 5a** illustrates the contour plot of wavelength-dependent birefringence of TE_{00} and TM_{00} for a range of waveguide widths. The dotted and solid lines in this contour plot represent the loci point of $|\Delta n| = 0.001$, followed by 0.01, 0.02, and 0.03, respectively. In this work, we are particularly interested to achieve the birefringence within $|\Delta n| \leq 0.001$. We find that this condition is feasibly achieved over a broad range of widths, w w and a large spectral range from 200 to 700 nm and 700 to 1800 nm, respectively. The optimum width is found within 200–300 nm that allows a zero-birefringence wavelength tunability in the range between 1200 and 1600 nm. This range is slightly narrower compared to the one obtained for the buried waveguide that is achievable within w $w = 375$ to 600 nm.[15] Introducing a thick slab, eventually, enables ultra-compact photonic device fabrication. On top of that, at an optimum width of 250 nm, it offers the largest tolerance of Δw Δw= 100 nm yielding wavelength tunability between 1190 and 1600 nm. This amount is broader by 135 nm compared to the buried waveguide that allows the wavelength tunability to range from 1360 to 1625 nm at w w= 376 nm as reported by Schollhammer.

We extend our analysis to observe the group index birefringence as the nonlinear properties and waveguide dispersion of the material are significantly raised on ultra-small cross-sectional waveguide.[13] These unique properties enable access to a wide range of nonlinear applications including broadband wavelength conversion,[20,29] entangled photon pair generation,[9] femtosecond pulse transmission,[30,31] and soliton generation.[32] Obtaining perfect group birefringence is crucial to minimize temporal walk-off and enhance the performance of nonlinear devices in waveguides. Deviating from this condition can reduce the temporal overlap between interacting pulses, resulting in lower conversion efficiency of nonlinear processes like second-harmonic generation, difference-frequency generation, and four-wave mixing. Additionally, temporal walk-off can contribute to pulse shape modification, causing pulse narrowing or broadening, as different spectral components of an optical pulse propagate at different velocities in the waveguide.[31] On the other hand, recent work has reported exploitation of the zero group birefringence effect to achieve the generation of Kerr solitons and coherent frequency

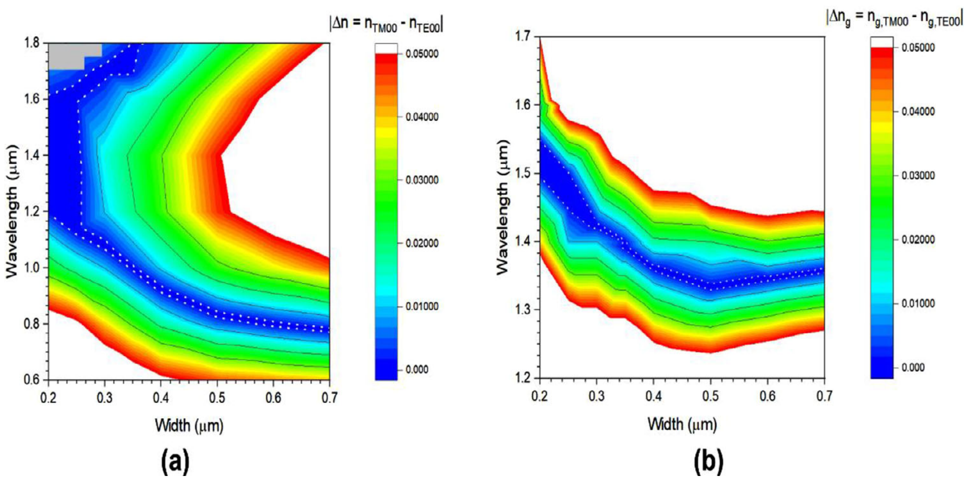

Figure 5. The influence of wavelength and width of core on a) effective indices, Δn and b) group indices, Δn_g of the fundamental mode, TE_{00} and TM_{00} . The harsh shape of the plot is due to the limited number of points, which results from the high computational load of the methods applied.

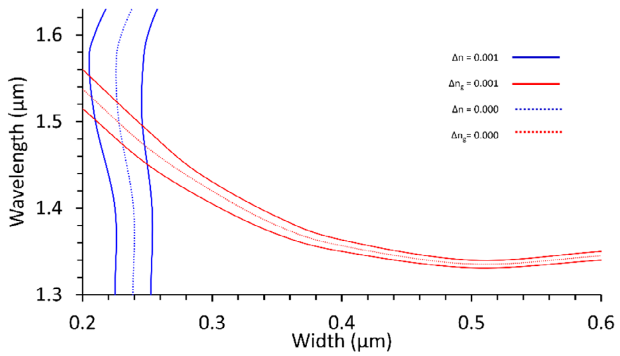

Figure 6. The overlap of the effective indices difference, Δn (blue) with group indices difference, Δn_g (red). The solid and dotted lines correspond to the loci at 0.001 and 0.000, respectively.

combs.[33] The group birefringence, Δn_g is quantified from the group index difference between TE_{00}, $n_{g,TE00}$ and TM_{00} ($n_{g,TM00}$). These group indices are derived from $n_g = n_{eff} - \lambda \frac{\Delta n_{eff}}{\Delta \lambda}$.

Figure 5b represents the contour plot of the computed group indices difference as a function of width, w w and wavelength, λ The blue region indicates $\Delta n_g \leq 0.01$. We find a gradual decrement of zero-group-birefringence wavelength as the width of the core increases. The dotted line indicates the group birefringence tolerance of $\Delta n_g \leq 0.001$. Such near-zero group birefringence is achievable over a large wavelength range from 1380 to 1580 nm. At $\lambda = 1495$ nm, the largest width tolerance, Δw of 75 nm is reported, yielding w w from 210 to 285 nm. On top of that, the largest wavelength spectral tolerance amount to $\Delta\lambda = 100$ nm is reported at $w = 250$ nm. At a specific waveguide length, L the pulse duration that can be generated by the devices directly from the relation of $\Delta\tau = L \, \Delta n_g \, / \, c$ where c corresponds to the speed of light in vacuum. In this case, a lengthy device and larger birefringence tolerance will maximize the pulse duration. Hence, a properly arranged waveguide length (˜in cm) as well as minimum birefringence tolerance of 0.001 enable ultrashort pulse propagation for various applications, including those at the single-photon level.[34]

We further analyze our findings by superimposing both effective and group indices contour plots for $\Delta n \leq 0.001$ and $\Delta n_g \leq 0.001$, respectively. From **Figure 6** it is found that the ZBC, $\Delta n \leq 0.001$ and walk-off-free operation $\Delta \ n_g = \ 0.00$ are achievable at w $w = 250$ nm at $\lambda = 1500$ nm. Compared to the buried structure reported in,[15] the rib waveguide offers a slightly longer operation wavelength at narrower width that is useful for propagating femtosecond pulses.

4. Conclusion

In this paper, the modal birefringence properties of an LNOI rib waveguide were presented for a broad wavelength range. It can be concluded that for a fixed wavelength, zero birefringence is achievable at a narrower width in rib waveguides compared to the buried and ridge structures. With a proper selection of the slab's thickness, the performance of the waveguide is improved significantly, and the waveguide can support a broad spectral range. Our results may be useful and relevant to a wide range of photonic applications, including nonlinear optics, and single-photon technologies. In nonlinear optics, modal birefringence can be exploited to control the polarization state of light as it propagates through a waveguide. For example, by carefully engineering the birefringence of a waveguide, it is possible to create devices such as polarization converters, which can convert a horizontally polarized input signal into a vertically polarized output signal or vice versa. These devices are important in applications such as telecommunications, where polarization control is critical to maintaining the fidelity of the transmitted signal. Apart from that, in single-photon technology, modal birefringence plays a key role in the design of devices that are used to manipulate and measure the polarization state of single photons. Additionally, polarization-entangled photon sources, which are important for quantum communication and computing applications, often rely on birefringence waveguides to generate and manipulate entangled photon pairs.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

lithium niobate on insulator, waveguide, wavelength tunability, zero birefringence

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